



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल



Slippers2Sat-2 (S2S-2): Power Budget

Kathmandu, Nepal.



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Power Generation Simulation



Power Generation Prediction using MATLAB (SSO)		
Parameters	Abbreviations/Units	Values
Solar Cell Efficiency	(cEff)	0.30
Solar Constant	(sPower, [W/m ²])	1353
Number of Exposed Panels for Each Side	nPY, nMY, nPZ, nMZ	4
Area per Panel of Satellite	(cArea, [m ²])	0.003
Generated Power	[mW]	1335
Total Energy Generated, A	[mWh]	2062
Energy Consumed by Blocking Diodes, B	[mWh]	310
BCR efficiency, C	[η_1]	0.8
Total Energy after BCR, D = (A-B)*C	[mWh]	1401.6
	20% margin	1121.28



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Normal Mode



BUS/ Mission	Electrical Characteristics			Active Period				Parameters			Burner	Telemetry			Mission			
	Voltage (V)	Current (mA)	Power (mW)	Period (sec)	Period (min)	Period (hr)	Repeat (@ 1 Orbit Cycle)	Total operating time in 1 Orbit (hr)	Energy Consumed (mWh)	Energy Consumed (Wh)		Beacon	HK-Data	DP-M	ADCS	CAM	EPDM	
EPS/OBC (without Flash write)	4.2	112.773	473.6466	5700	95	1.583333333	1	1.583333333	749.94045	0.74994045	749.94045	749.94045	749.94045	749.94045	749.94045	749.94045	749.94045	
EPS/OBC (Flash write)	4.2	176.5	741.3	20	0.333333333	0.0055555555	63	0.35	259.455	0.259455	OFF	OFF	259.455	259.455	259.455	259.455	259.455	
COM TX (Beacon)	3.3, 4 (MCU, Power Amplifier)	22.5, 950	3874.25	1	0.0166666667	0.0002777777	47	0.0130555556	50.58048611	0.0505804861	OFF	50.58048611	OFF	OFF	OFF	OFF	OFF	
COM TX (Downlink)	3.3, 4	22.5, 950	3874.25	120	2	0.0333333333	1	0.0333333333	129.1416667	0.1291416667	OFF	OFF	129.1416667	OFF	OFF	OFF	OFF	
COM RX (Uplink)	3.3	4.8	15.84	5700	95	1.583333333	1	1.583333333	25.08	0.02508	OFF	25.08	25.08	OFF	25.08	25.08	25.08	
DP-M TX (Downlink)	3.3, 4	22.5, 950	3874.25	120	2	0.0333333333	1	0.0333333333	129.1416667	0.1291416667	OFF	OFF	OFF	129.1416667	OFF	OFF	OFF	
DP-M RX (Uplink)	3.3	4.8	15.84	5700	95	1.583333333	1	1.583333333	25.08	0.02508	OFF	OFF	OFF	25.08	OFF	OFF	OFF	
ADCS	3.3, 5 (Sensors/MCU U, MTQ)	0.6, 3.4, 28.6, 98	892.38	10	0.1666666667	0.0027777777	1	0.0027777777	2.478833333	0.0024788333	OFF	OFF	OFF	OFF	2.478833333	OFF	OFF	
ADCS	3.3 (MCU only)	28.6	94.38	1	0.0166666667	0.0002777777	1	0.0002777777	0.0262166667	0.0000262166	OFF	OFF	OFF	OFF	OFF	0.0262166667	OFF	
CAM	5	452.0	2260	10	0.1666666667	0.0027777777	1	0.0027777777	6.277166667	0.0062771666	OFF	OFF	OFF	OFF	OFF	6.277166667	OFF	
EPDM	3.3	81	267.3	1800	30	0.5	1	0.5	133.65	0.13365	OFF	OFF	OFF	OFF	OFF	OFF	133.65	
Burner Circuit	3.6	2200	7920	15	0.25	0.0041666666	1	0.0041666666	33	0.033	33	OFF	OFF	OFF	OFF	OFF	OFF	
GPS			60	5700	95	1.583333333	1	1.583333333	95	0.095	OFF	0.095	0.095	0.095	0.095	0.095	0.095	
Total Energy Consumption (mWh)											782.940	825.696	1163.712	1163.712	1037.0492833	1040.848	1168.220	



BUS/ Mission	Electrical Characteristics			Active Period				Parameters			Safe Mode
	Voltage (V)	Current (mA)	Power (mW)	Period (sec)	Period (min)	Period (hr)	Repeat (@ 1 Orbit Cycle)	Total operating time in 1 Orbit (hr)	Energy Consumed (mWh)	Energy Consumed (Wh)	Beacon
EPS/OBC (without Flash write)	4.2	112.773	473.6466	5700	95	1.583333333	1	1.583333333	749.94045	0.74994045	749.94045
EPS/OBC (Flash write)	4.2	176.5	741.3	20	95	1.583333333	1	1.583333333	1173.725	1.173725	OFF
COM TX (Beacon)	3.3, 4 (MCU, Power Amplifier)	22.5, 950	3874.25	1	0.01666666667	0.0002777777778	46	0.01277777778	49.50430556	0.04950430556	49.50430556
COM TX (Downlink)	3.3, 4	22.5, 950	3874.25	120	2	0.03333333333	1	0.03333333333	129.1416667	0.1291416667	OFF
COM RX (Uplink)	3.3	4.8	15.84	5700	95	1.583333333	1	1.583333333	25.08	0.02508	OFF
DP-M TX (Downlink)	3.3, 4	22.5, 950	3874.25	120	2	0.03333333333	1	0.03333333333	129.1416667	0.1291416667	OFF
DP-M RX (Uplink)	3.3	4.8	15.84	5700	95	1.583333333	1	1.583333333	25.08	0.02508	OFF
ADCS	3.3, 5 (Sensors/MCU, MTQ)	0.6, 3.4, 28.6, 98	892.38	10	0.1666666667	0.002777777778	1	0.002777777778	2.478833333	0.002478833333	OFF
ADCS	3.3 (MCU only)	28.6	94.38	1	0.01666666667	0.0002777777778	1	0.0002777777778	0.02621666667	0.00002621666667	OFF
CAM	5	452.0	2260	10	0.1666666667	0.002777777778	1	0.002777777778	6.277166667	0.006277166667	OFF
EPDM	3.3	81	267.3	1800	30	0.5	1	0.5	133.65	0.13365	OFF
Burner Circuit	3.6	2200	7920	15	0.25	0.004166666667	1	0.004166666667	33	0.033	OFF
GPS			60	5700	95	1.583333333	1	1.583333333	95	0.095	OFF
Total Energy Consumption (mWh)											799.445